

# Archishmat (Archis) Gore

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## SUMMARY

Engineer, founder, and security researcher with twenty years building systems at scale — at Amazon, Meta, and eight years as co-founder and CTO of Polyverse, where I shipped the first moving-target-defense Linux distribution adopted by the US Navy. The through-line in my work is diversity as a defense: closure-based randomization applied at every layer of the stack — bytecode, calling convention, kernel ABI, custom ISA — so that the cost of breaking many targets is  $N \times$  the cost of breaking one. Captured in USPTO 10,733,303, which I pledged to the public domain.

## EXPERIENCE

### Meta Platforms — Engineering Leader, Facebook ML Reliability

Seattle, WA · 2022 – present

- Responsible for all ML reliability for Facebook.com — availability, stability and quality. Improved AI model reliability from 99% → 99.99% for the primary user-facing inference path.
- Eliminated several categories of infrastructure failure, cutting operational load 100× — an estimated 2,000+ engineering-years recovered across affected teams.
- Built business-driven ROI metrics for infrastructure reliability, now used by multiple orgs to prioritize multi-quarter roadmaps.
- Leading the company-wide "Mass Model Administration" initiative — governing 100× more models in production with no headcount growth.

### Polyverse Corporation — Co-founder and Chief Technology Officer

Bellevue, WA · 2015 – 2022

- Built the engineering org and the IP from zero. Drove the company from patent filing to a commercial product line and government adoption.
- Invented and shipped polymorphic Linux: the only complete OS distribution where every build's binary ABI is unique. In active production with the US Navy. No other vendor has shipped anything comparable.
- Container-cycling defense (DoD-adopted): when an exploit lands, no more than ~0.1% of the application surface is exposed at any time because the rest is cycling out from under it.
- Polymorphing Build Farm — a complete supply-chain-in-a-box that rebuilds firmware, bootloader, kernel, and userspace libraries from source in a hermetically sealed environment with a fresh per-build seed.
- Partnerships with MITRE, DISA, Red Hat, Oracle on supply-chain and binary diversity. Contributing member of OpenSSF Supply Chain Security WG and the Bytecode Alliance.
- Authored the Encrypted Execution thesis (encrypted-execution.com) and the patent (USPTO 10,733,303) underlying the product family.

### Amazon.com — Head of Global Search Reliability

Seattle, WA · 2011 – 2015

- Owned reliability for one of the highest-revenue surfaces on the public internet.
- Drove Amazon.com homepage and search availability from 99.9% (~6 hours/year of downtime) to 99.99999% (~7 seconds/year).
- Invented a latency-to-mission-accomplishment metric that surfaced hidden customer-impact latency hot spots; remediation produced ~\$200M annual revenue improvement.

- Migrated the Product Data Service from a monolithic Java codebase to a serverless tier-1 service; transferred business-rule ownership to the respective product teams.

## **Microsoft Corporation — Software Engineer (CSA Office · Azure · Bing)**

*Redmond, WA · 2005 – 2011*

- Won the India-wide Code4Bill competition; one-year internship followed, with competition answers integrated into Bing.
- Core member of Ray Ozzie's elite development team that shipped Azure Service Fabric v1.0, Live Mesh, and other foundational cloud-era projects.
- Led the team that shipped Microsoft's News Reader app, a direct competitor to Google Reader.
- Primary sole developer of Live Mesh for Windows Phone.

## **SELECTED OPEN SOURCE AND CURRENT RESEARCH**

- encrypted-linux — research distro extending polymorphic Linux to a full scrambling-GCC toolchain. Per-build seeded ABI diversity for every binary on the system. Active.
- WineTone (tone.wine) — a wine recommender that calibrates to each user's personal vocabulary. A demonstration of user-context recommendation: each user has personal context on how they use words.
- zerotect — exploit detector that uses polymorphism to surface crash-induced side channels from polymorphic builds.
- EnVisen — ROP gadget finder and analyser in pure JavaScript.
- rmesg (crates.io/crates/rmesg) — dmesg in Rust, with a streaming library API. 13,000+ downloads/month.
- polyscripted-wordpress — reference implementation of per-build lexer dialects for PHP / WordPress.

## **TECHNICAL DEPTH**

**Programming languages:** Rust, C, C++, x86/ARM assembly, Go, Python, Java, Erlang, JavaScript/TypeScript.

**Compiler / toolchain:** LLVM, GCC, JVM, V8/Node.js. Author of the scrambling-GCC fork underlying polymorphic Linux.

**Architectures:** x86\_64, ARM (including Apple Silicon), RISC-V, custom-ISA work for encrypted-execution research.

**Operating systems:** Linux (contributor), Windows, RTOS optimization, hypervisor surfaces.

**Distributed systems:** Owned tier-0 surfaces at Amazon scale and Meta scale; comfortable with consensus, sharding, regional failover, hot-cold patterns, async messaging.

**AI / ML:** Building production reliability surface for Facebook's user-facing inference at Meta; researching personalization and embedding calibration at WineTone.

## **SELECTED INVENTIONS AND IP**

- USPTO 10,733,303 — Polymorphic compilation systems. Granted 2020, pledged to the public domain. Underpins polyscripting, polymorphic Linux, and the encrypted-execution thesis.
- Encrypted Execution whitepaper — formal framing of closure-based diversity as a defense primitive. encrypted-execution.com.
- Polymorphic Linux — the only complete Linux distribution where every build is its own ABI. Shipped at Polyverse 2015–2022; in active production with US Navy and other government customers.
- Code4Bill (India national programming competition) — winner. Answers adopted into the production Bing search stack.

## **EDUCATION**

M.S., Computer Science — University of Pune, India

B.S., Computer Science — University of Pune, India

